Clustering Techniques

K-means Clustering

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DAI-101

Topics Covered

- Various Loss functions in ML; Gradient Descent and Stochastic Gradient Descent; Bias Variance Trade-off, Overfitting and Under fitting
- Logistic Regression with examples, Softmax regression
- Bayes Theorem with posterior and Prior, Naïve Bayes Classifier
- Decision Trees, Entropy, Gini index, Examples for classification and regression
- Basic notion of unsupervised learning, Clustering, K means, Elbow method, DB score, metrics like Sillehoute score in clustering
- Hierarchical Clustering and DBScan
- Gaussian Mixture Models with Expectation Maximization Algorithm
- Artificial Neural Networks, MNIST data set [if time permits]

Additional Content

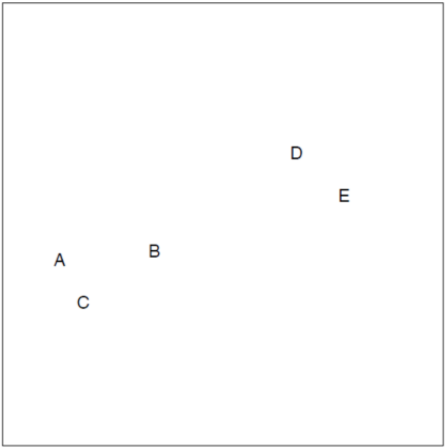
- Prof. Sanjeev Kumar – https://drive.google.com/drive/folders/1gMGMvvAay98fF6xLgTj6QyX4wCm_GUNB?usp=sharing
- Prof. Sumit Kumar Yadav – https://tinyurl.com/iitrdai101

Last Lecture

- K-means clustering
- Elbow method, DB index
- Silhouette Score

- K-mean clustering.
    - pre-specify the number of clusters K.
    - Depends on the random initialization.
- Hierarchical clustering is an alternative approach
    - Agglomerative (fine to coarse) hierarchical clustering.
    - Devisive (coarse to fine) hierarchical clustering

Agglomerative hierarchical clustering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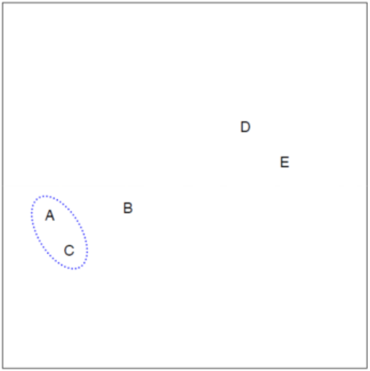
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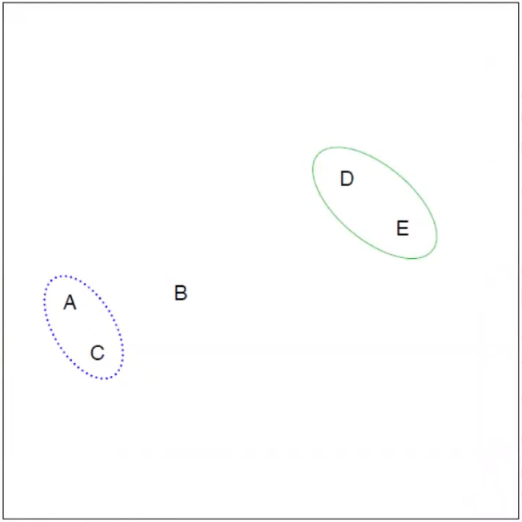
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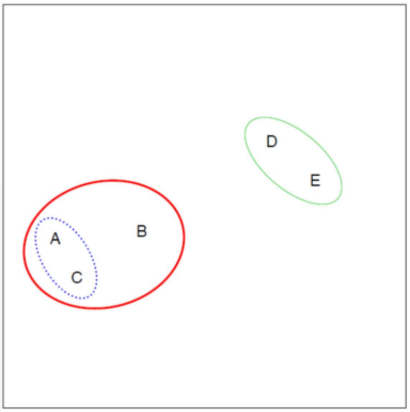
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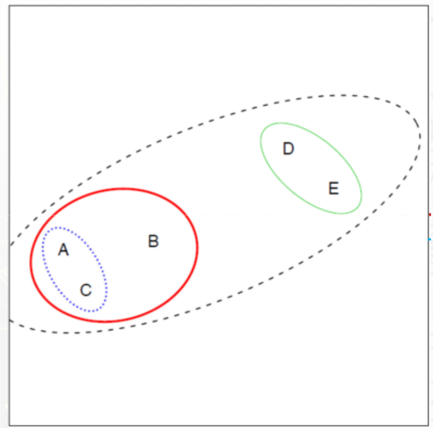
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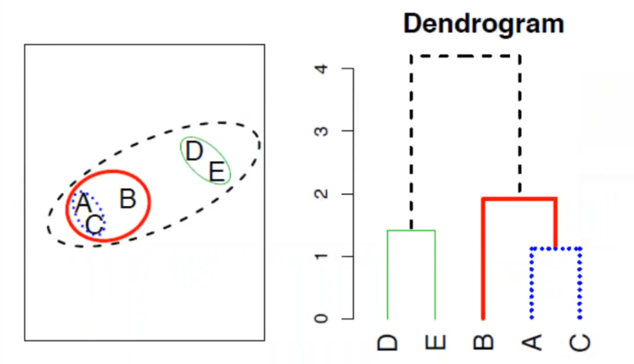
Dendrogram



y-axis on dendrogram is (proportional to) the distance between the clusters that got merged at that step

Steps in Algorithm

Step 1 Start with each point in its own cluster.

Step 2 Identify the two closest clusters. Merge them.

Step 3 Repeat until all points are in a single cluster.

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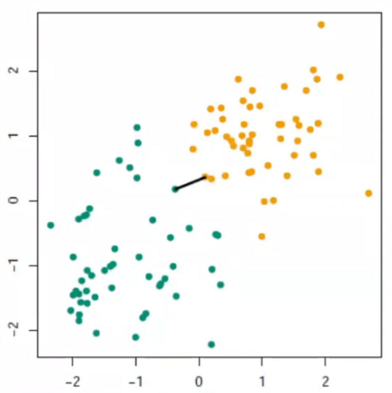
> How to measure the distance between clusters?

- Single linkage: Minimal inter-cluster dissimilarity.
- Complete linkage: Maximal inter-cluster dissimilarity.
- Average linkage: Mean inter-cluster dissimilarity.
- Centroid linkage: Centroid inter-cluster dissimilarity.

Single Linkage

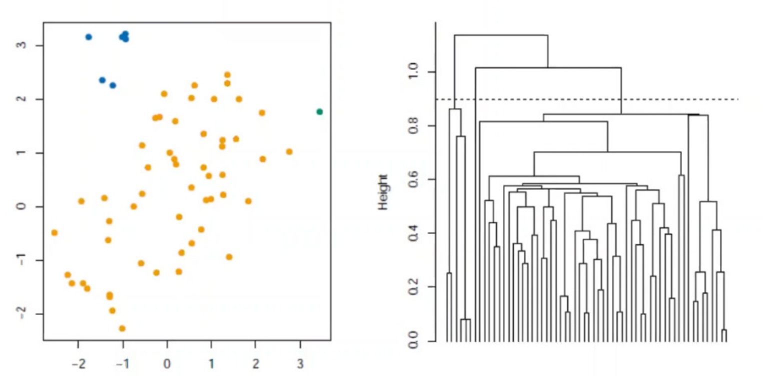
The distance between G, H is the smallest distance between two points in different groups.

$$d_{single}(G, H) = \min_{i \in G,\ j \in H} d(x_i, x_j)$$



Single Linkage

Here $n = 60, x_i = \mathbb{R}^2, d_{ij} = ||x_i - x_j||_2$. Cutting the tree at $h = 0.9$ gives the clustering assignments marked by colors.

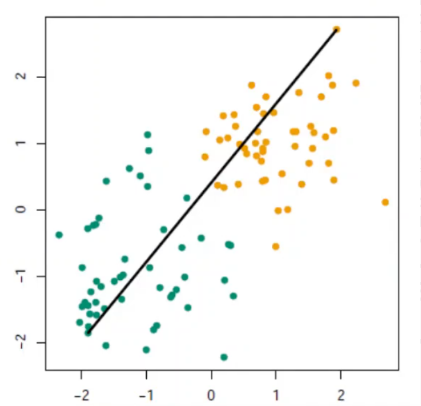


Cut interpretation: For each x_i, there is another point x_j in its cluster such that $d(x_i, x_j) \leq 0.9$

Complete linkage

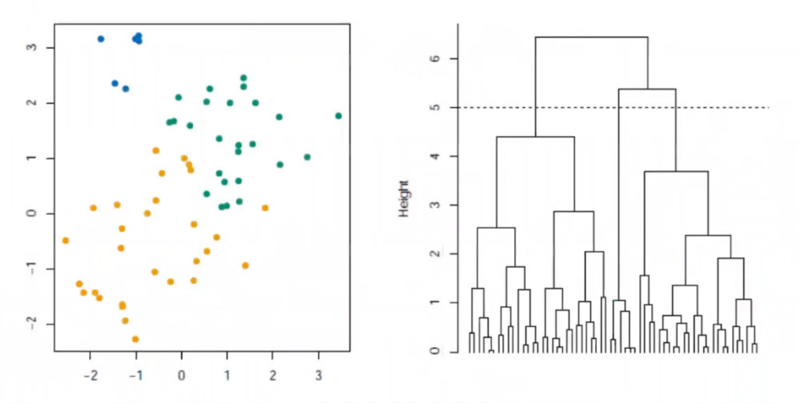
The distance between G, H is the largest distance between two points in different groups.

$$d_{complete}(G, H) = \max_{i \in G,\ j \in H} d(x_i, x_j)$$



Complete Linkage

Here $n = 60, x_i = \mathbb{R}^2, d_{ij} = ||x_i - x_j||_2$. Cutting the tree at $h = 5$ gives the clustering assignments marked by colors.



Cut interpretation: For each x_i, every other x_j in its cluster satisfies that $d(x_i, x_j) \le 5$

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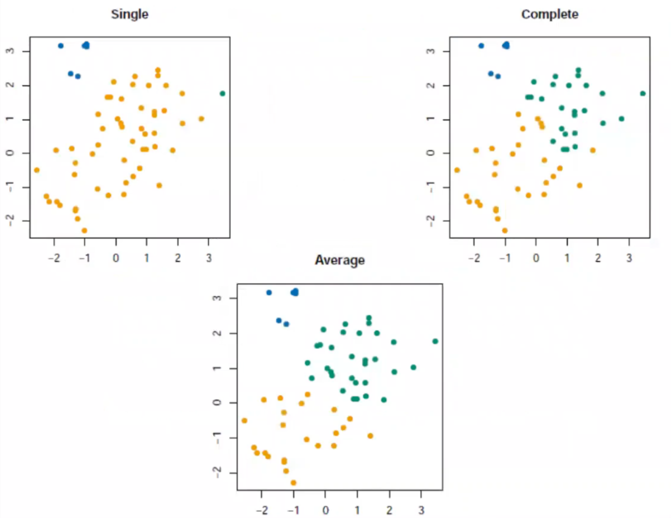
- Average linkage tries to strike a balance.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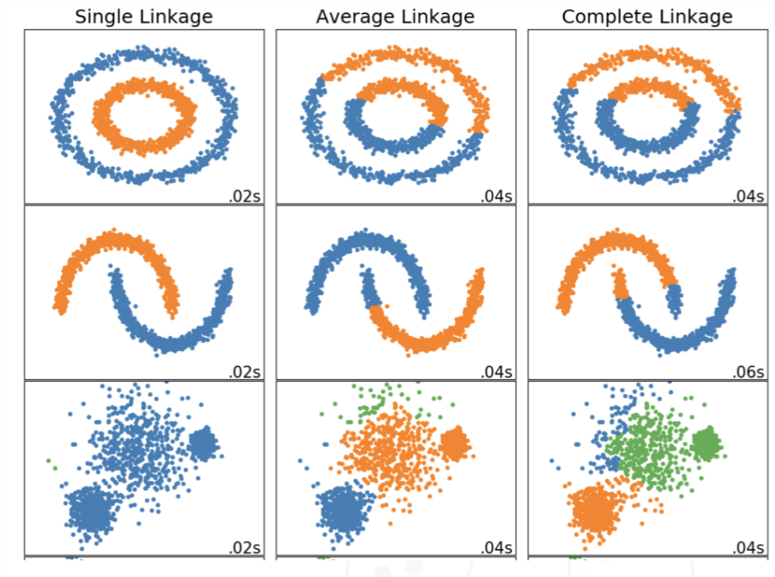
Single Linkage
Average Linkage
Complete Linkage
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demo

Thank you!

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